

# ActInf Textbook Group ~ Cohort 4 ~ Meeting 11

## (Chapter 5 part 1)

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Duration: 59:18 | Uploaded: Unknown Transcript method: auto\_caption

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all right welcome thank you everyone it's August 15 23 and word cohort 4 in our first discussion of chapter five so soon we will screen share but first any remarks that anybody wants to make on chapter five just anything that stayed with them after reading it or feelings that they had or anything else about chapter five I will say uh I found this to convert and conversely to chapter four being rather mathematically against this one is a little bit more of neurobiologically dense at least for those who maybe like are new to um thinking about the relationship between this and areas of the brain but I did appreciate in this chapter there was much more on Precision which we didn't get too much before um even though it was brought up many a Time preceding chapters and I really appreciated the relationship between uh that was made between different forms of precision and different neurotransmitters neuromodulation we found that to be very interesting and helpful and maybe kind of reinforcing the idea of like they're not just probability distributions here but also uh Precision of each or like confidence in each probability distributions so my big takeaway from this chapter cool thank you anyone else is there any thoughts on five any other thoughts or any any question that somebody wants to lead with or we can just explore the chapter and look at some written questions um I did I did want to ask uh what everyone makes of uh the relationship made here I mean in preceding chapters we're talking a lot about um you know kind of the differences between viewing variables as categorical versus continuous and it's a big kind of bifurcation throughout many of these chapters like we have continuous models versus um you know Palm DPS and um just different ways of looking at things but it at one point in this chapter they point to the idea that um there's actually a big relationship a connection between the two that and maybe what's going on in the upper levels of say the the brain cortical regions and so on is that they're working with categorical variables but in a way it's not simply categorical variables but rather these are like um looking at discrete trajectories that the continuous variables could take that was a nice way of maybe putting making some kind of

thread between the two ways of looking variables that is there's still a continuous aspect within the categorical States um it's just it's looking at it as a discrete trajectory I didn't know if anyone made anything of that found it to be useful for anything or anything they're working oh thanks um Darius than anyone else yeah it's not so much a comment on the previous Point um about discrete versus continuous variables I wanted to ask really a question more uh more concerned with the mechanisms behind prediction coding at least as it's embodied in the brain so you I gotta understand this idea of sort of the shuttling up of prediction errors and the shuttling down of expectations I guess Daniel my question is we kind of just take this idea that prediction errors arise for granted um something I've been trying to kind of wrap my head around is to what degree is that a kind of just um immediate mismatch between the priors at one layer and the incoming sense data or is it that those there's some Bayesian inference required to even compute the prediction error so you know I have a prior  $x_i$  compute variable base to work out the probability of my prior given  $y$  is that in a sense that distinction between the prior and the posterior is that what is constituting surprise or is it just a pure sense data coming up the chain which then is discordant with my prior  $X$  if that if any of that makes sense yeah what a lot of uh a lot of angles of this just first very briefly yeah this uh hierarchical hybrid type of model where the higher levels of the system are discretized and then the lower levels are kind of like where the rubber hits the road is being more continuous that's a motif that comes up like again and again for example in Ryan Smith's um at all's uh work live stream 46 um active inference does not contradict folk psychology this is a common Motif and a big reason is that counterfactuals are not really so clear in a continuous setting like if there's um three discrete choices you can really quickly talk about the tree of all choices but if it's just like a cone of options you could take like an infinite amount of paths because technically even paths that are only one you know infinitesimal apart in the continuous space are still distinct trajectories so in the continuous time you don't really get a kind of explicit planning or explicit counterfactual and so in order to deal with cognitive decisions which which often are um categorical or discrete or deal with like explicit symbolic actuals then you need some kind of discretization and yes the good news is that they work well together and that gets Revisited in in practice when all of the difference neural models are going to get brought together and shown that there's no issue with the continuous uh model here in the discrete model up here and then also comes back in chapter eight okay about mechanisms of predictive coding in the brain well it's interesting to look at the first words in this section chapter 4 gave us a lot of general information about variational inference about generative

models um essentially following from free energy principle in this chapter we focus on the process theories which are explicitly about a system of Interest so whereas a principle can't be falsified or not it's just it's something that is it um in the realm of decisions that you take towards modeling process theories are more in line with scientific hypotheses they're actually like kind of living and dying with empirical evidence but the amount of empirical evidence for any given statistical model of brain has nothing to do with um whether the principle is valid or not now that doesn't mean it it obviously is it's good when we see empirical support for process theories under the principle but there's no piece of evidence that could be found I mean maybe this is an overstatement that would validate a principle because principles aren't validated by evidence but nor are they rejected by evidence however it's super interesting to connect them to specific systems so we're going to be applying generative models to the brain Bayesian brain plus action and one of the kinds of models but but not the only kind of generative model that exists for the brain is like the hierarchical predictive coding architecture and so in that architecture um which arose from video compression but then was later found to capture a lot of biological phenomena there's an alternating um stack of error units and kind of estimator units so that the different levels of estimator units are sending um ascending error signals and descending predictions so Darius you're you're asking is that mismatch immediate or is it computed through a Bayesian update I mean is that the question yes basically well the trivial deflationary answer is Bayesian updates happen in Bayesian statistics so if you want to make a statistical model that calculates an exact mismatch or if you want to have it do basing updating that's about the map not the territory so we're already lifted away from what the system's already doing um in this chapter in the motor unit here it is just a direct mismatch descending predictions of the of the mean incoming proprioception and it's just a simple differential but that simple but there's no reason why you couldn't make it slightly different um like you could have I mean you you could do anything with Legos that doesn't mean it would be more effective or or give unique explanations or predictions but in its simplest form which is embodied here it is just a differential denominated in units of the measurement could you do a Bayesian update I mean could uh could this variable what what would that do or what would we get by having this do a Bayesian update I mean I sense the difference between the two models would be one you have predictions coming down the stream and you get the mismatch coming from what the higher levels are expecting and it's kind of meeting at this point where the prediction error is occurring the other one is that the prediction occurring is just happening within a level where there is a prior that prior is then

getting updated given evidence and there's going to be a mismatch between the posterior and the prior so there's that's all happening within the level the prior of the level and the posterior of the level whereas the other way of thinking about it is the kind of Prior of a higher level the proximate level and that's shuffling down you get prediction now shutting up but the kind of level where it's all happening is just a meeting point under that model yeah I see what you're saying so like if if this were instead of just a simple simple differential If This Were Like a Coleman filter so it had its own inertia and it was yeah it was doing it it was getting input and then it was doing some like filtering just like common filter and then it was sending some signal back up yeah there's an infinite amount of model architectures you could you could have any you know you could compose it however you want in principle and no piece of evidence can support or reject the in-principle construction or composition so there are definitely some interesting properties of that and then this is where we get to the process theories so then we look at specific measurements or or cognitive or behavioral phenomena in biological systems for example and then we seek to compose something that um captures some of the phenomena that we want to capture recognizing that maps and territories uh are not the same thing and then you have a whole portfolio of models and you compare them in terms of their um accuracy explainability adequacy all of these standard statistical modeling methods then you write the paper with all the compared models and you publish the code then other people continue developing that's kind of more normal science than it can sometimes seem but the earlier chapters talked a lot about what what could be done I mean and other literature goes goes beyond that and then they're they're not saying this is how it is they're um it's not the final word on process theories for active inference it's simply the interpretation consistent with available evidence so they're just saying here's one model architecture that we're using for Unique explanations and predictions in these different neural systems now those specific model architectures are at the mercy of data like if someone said I think these three factors contribute to this health outcome that's a hypothesis and then someone's good thing oh no I think it's just these two and someone else says I think there's a fourth Factor those are all going to be compared just like regular scientific modeling whereas none of those discussions are going to be like well the the three-factor model doesn't fit the health condition so linear regressions are incorrect that's like a category error but now in chapter five we've kind of crossed the river from the General High Road low road to now I guess I don't know specific cars on the high road the low road something like that so this gives us a very different like flavor and deployment of active because we're so directly talking about and these are the

ones that are in contact with the empirical any other thoughts or questions on this okay well chapter five is going to um go through three neural systems from the mammalian nervous system foreign and in the end it's going to be kind of a frontal cortex a midbrain region and a spinal cord region and the chapter is going to introduce these three different models which essentially um cover some connected the distinct cognitive phenomena and those three systems are going to be considered separately and then it all gets built up to figure 5.5 which composes these three systems and starts getting you towards that whole iguana so this is like how how um functional composition for these kinds of generative models works we're not going to go into the whole category Theory elements of compositionality here but it's just encouraging to know that the generative models that somebody makes um are are are able to be synthesized and and play nicely together so you could also compose one of the the brain and you could compose one of a computer and then that would be your human computer interface or you could have two humans and then that would be your communication system and um so that that kind of model compositionality is a is a very nice feature that chapter five makes kind of an understated argument in support of hi Ali 5.2 micro circuit semesters any any overall thoughts or anyone want to like give a summary on what they read or what they were left with from this section um just to comments uh and this kind of follows from something we briefly discussed last week in chapter four and one of the meetings but I found it very interesting the idea that that they're kind of like asymmetrical frequencies implied in this 5.2 section um that that you can have uh like non-linear functions in the lower level Computing uh you know errors related to incoming sensory information versus a prediction um and that's synced up in the in a narrow Imaging that seems to be commonly associated with like a gamma band frequency versus uh the descending predictions that are made which tend to be at like a slower rate like uh somewhere around Theta I suppose and um so it's just it felt like this was giving a little bit more concreteness to predictive coding and saying like there are differences in frequencies there are various functions or differences here it's not as simple as um taking a simple Vector of the predictions from the upper layer and the sensory information like and just doing some simple subtraction right like the actual minus the predicted it's rather that there there's different speeds of information movement here and different components different parts and different ways you could potentially model it awesome yeah and one short note a lot all out there and bring in to the notes um the reactive message passing Paradigm which I I hope we'll hear more about from Dimitri and Bert in the Symposium um these are computational implementations where different nodes in the graph

can update at different frequencies so it's perfectly amenable to that Darius yes I mean it's um it's in this section so it's page 87 yes yeah you were just there if you go down a little bit yeah this is that where it is it's above the middle schematic oh then yes perfect so if you just live with that um I think this is exactly what I was talking about which is it says here um in the sort of the middle part of the bottom paragraph you're sending output from layer 3 superficial pyramidal cells blah blah blah so the that layer is shooting prediction there up then you get the descending input representative prediction from the higher level presumably and while descending output is a prediction for the lower level so I think that is kind of maybe worth unpacking because it's I mean there seems to be a lot going on um and a lot of predictions going down and prediction error going up so it would be good just to get some clarity on exactly where all this is happening and you know what layers we're exactly talking about here good good point there's a few things so first I'll bring in one one blog post that kind of contextualizes like why are we focusing on this cortical architecture at all um Jeff Hawkins has a book and I guess some startup or something related to this cortical architecture so a lot of people have um whether they've looked at the gross neuroanatomy or they've looked more into the finer scale neuroanatomy a lot of people have centered on the the cortical regions of the brain which it's in the mammalian frontal part of the brain and it's the part that's wrinkly and on the outside that that type of tissue organization supports different cognitive phenomena that people are interested in of course not to say to other regions of the brain are not also doing different cognitive phenomena but that for certain kinds of cognition understanding the connections within and across the stereotypical units of the cortex called columns this has been a Super Active area of research now a really important note is that while there are six layers so-called six layers in this in the cortex these six layers are histologically defined basically so they're defined on tissue properties like what you can see under the microscope and like gene expression things like that this is not a six layer sandwich with prediction error prediction error six times so this is not a six level hierarchical model this is actually a uh but now this does have a hierarchical modeling but if you look at the subscripts or the superscripts we see  $I$  and  $I + 1$ . so actually these six histological layers together are composing like kind a single multi-scale jump and then the idea is that across cortical units because they also signal laterally not just within the column higher nested models could be constructed cognitively but this is not like a six layer nested model it just happens to be the case that this tissue has these six laminar histological layers um okay so the the layers with the numerals these are these are just anatomically defined these are not statistically defined at all

and the on the left the anatomy of this column stereotypical column is shown and each of these nodes represents like a cell population it's not necessarily just a single cell but perhaps like a group of cells that have some similarities and like the spdp s s i i these are like different cell populations and this on the left is showing their empirical anatomical connections that's not to say that's all of the relevant connections just like there's a subway map and then also there's footpath so there's bioelectricity there's neural hormones there's a lot more in play but this is just the wiring associated with these cell populations now as common in computational Neuroscience a given single cell or population of cells is going to be associated with some variable on a statistical model and here we basically have um two main letters so first the tilde means Through Time the two main letters here are the MU and the um the e the MU is like the mean estimate and the e is um the error signal I'm I'm a little bit not sure why G is being discussed here rather than mu olly do you have a thought on this uh well uh yeah actually I guess that's because uh you see based on um songs framework you see uh all are all the brains needs or uh the the way that uh the signals need to be uh passed through both uh through these uh this bi-directional path that they don't need to be satisfied all at once so obviously their prior prioritized by uh by a decision-making triangle uh in which uh the immediate current needs uh or in other words uh the residual predict prediction errors they Converge on uh the as Psalms describe in his book uh periquidactyl gray and they're ranked in relation to uh the uh the current needs and opportunities uh in and and this opportunity is manifested in a in the form of a kind of two-dimensional saliency map uh in the superior colliculi so uh this um prioritization uh triggers a kind of conditioned action programs uh which uh in turn unfolds in expected uh context over a kind of deep hierarchy of predictions or in other words the general model of the um of the forebrain so these actions uh they're generated by a prioritized uh affects that are generalized that are generated by prioritized ethics they're somehow voluntary uh in other words they're subject to the immediate Here and Now choices right uh kind of pre-established programs or pre-established algorithms and on the other hand these choices are kind of felt in uh if we want to talk about the higher order um the higher order elements of the brain they're kind of felt in the extraceptive Consciousness which again contextualizes this kind of affect and uh they're they're made on the basis of this fluctuating uh Precision waiting uh in other words uh arousal modulation um and postsynaptic gain of the incoming error signals uh that are on the other hand rendered silent by these prioritized needs and they're uh they're also buffered in working memory uh with the aim of minimizing uncertainty or maximizing confidence uh which also correlates with the current prediction as to how that

particular need or particular I don't know desire to minimize the prediction error needs to be met so briefly what this amounts to uh is a kind of a bit unsettling which kind of says that the higher order elements of the of the brain or the Consciousness would only be required to mediate mismatches between the internal and external States and the more adaptively and predictably reliable our model becomes the more our Behavior will become automated uh in other words operating below the level of conscious experience yeah I guess that balance between the uh I mean the Precision making of the gnd E comes from uh or at least can be explained from uh psalms's framework in this way oh awesome but there's no G in this figure well here's G here's the full descending G so just be just just like um you know the knee bones connected to the leg bone and all of that ascending and descending are relative to obviously which node we're talking about so the ultimate descending um prediction again in a in a a standing person the ultimate descending prediction is going to be like the the main big output of this model and that's what's going to uh it's not shown here but it's it's what is the the full output of this whole column is this G I mean there there's so many oh yeah I'll go for it no I just wanted to add another uh Point here uh which uh you see these kinds of Errors uh well the word error can somehow um connote negative meet uh I believe in the free energy principle framework or active inference framework or at least Mark zomes's interpretation of it uh that's exactly what our Consciousness feeds on so um in other words uh the surprisal uh I mean uh the Consciousness as I said is only needed uh I mean roughly speaking in situations where surprisal needs to be minimized uh if no surprise though there would be not much need for Consciousness or to put it in a different terms at Consciousness is a kind of process that seeks to somehow overcome itself so these kinds of error are essential for uh for any higher order activity of the brain be it I mean Consciousness decision making or any kind of higher order activity so it would never be exactly uh equated to zero because that would basically mean uh to go unconscious and to not have any higher order activity uh in the brain so this also maps onto annual Seth's Theory Of Consciousness as to uh as a kind of process that um seeks to minimize the surprisal yeah also error um yeah like you said Eric can seem like it's a bad thing but it's like a set point and half the time your error is going to be well in a calibrated system half the time the error is positive and half the time the error is negative if you're spending more time if ninety percent of the time your estimates are too high in a galaxy in the world then that would be a signal to update there's a lot more detail to go into with the exact mechanics here um including the the difference between the Middle version and then the version on the right which is more connected with the notation that we saw from the pomdp

um but this is kind of like a um it's like a for some reason didn't give a fancy iced coffee percolator but something is coming in and this is like a percolation mechanism and there's a there's a descending from  $I$  plus one that's the  $G$  here prediction from the higher level and then there's the um ascending uh differential this isn't the raw signal data coming in this is already the differential coming up and then those are able to meet through a statistical mechanism that also recapitulates but but not exactly I mean it's not like it's this five and three it's it's it's not exactly intended to to be but this is the result of the this is a falsifiable process Theory that then somebody can explore like well what statistically happens when this when you transiently suppress this exact connection and then the process of model building continues Darius so I asked whether the  $I$  in this um is that referring to the tilde is it referring to time where  $I$  plus one is uh future or past predictions and whatever or is it actual is it part that is it relating to the level of the hierarchy correct  $I$  is the  $i$ th level of the hierarchy  $I$  plus one is a higher level the hierarchy tilde means Through Time now they're going to abstract a little bit from this six layer showing all the nodes so here we still have one two three four five six with figure five or I mean uh with a uh layer four um being shown now it's being clarified how things falling in my room now it's um being clarified how here we have  $I$  minus 1  $I$  and  $I$  plus one so these are three cortical columns that are next to each other and now those three cortical columns are able to actually instantiate a three layer nested model so it's almost like we get the vertical stacking through horizontal uh concatenation or composition where the outputs of one column can be errors like here from from  $I$  um and also notice that  $I$  that they're not bound within a column here's  $I$  minus one and then the error unit of  $I$  then the mean and the error units of  $I$  and then the error units of  $I$  plus one and then the highest nested layer on the right [Music] which is uh only sending descending predictions and getting the highest level ascending predictions yeah I'm just a bit closed with what you just said uh in terms of you know uh what  $I$   $I$  minus one and  $I$  plus one mean here so formula works you know three different time steps uh within the the tilde uh you know big Vector of of states uh and not you know a communication between different columns yeah the the  $I$  the superscript is not time subscript is used for time but here this is referring to three nested levels so let's just say this cortical column we're guessing the hundreds place this one where the tens place and this were the ones place so together this is a three layer nested that's um doing predictive processing on the three-digit number but time is not shown here it's just saying tilde it's all happening through and here the  $I$  superscripts are related to the level of the hierarchical model Darius sorry I don't know why it keeps going into clapping I do mean to clap everyone's making excellent contributions

um um it's so just to summarize it seems like there's hierarchy both Within and across so if you go back up to that it so the way you spoke of that before is this is a free layered nested hierarchy but we also have superficial and deep down the vertical what I mean it's a very general question but why is this computationally efficient to have both vertical and horizontal hierarchies why can't we just shuttle predictions and prediction error across one dimension yeah well I I stated this though it has six tissue layers is really like is this is like a minimal multi-scale model um you could make a model you could make a predictive processing hierarchy that's like a skyscraper and just simply has three on top but it just turns out that this is actually capturing the structure of the brain so I guess what I mean is what we what do we what's what's gained by I I you know it makes complete sense I mean whether it's vertical horizontal I think it's a moot point because it's just it's just way of I guess picture I guess but what's gained by having two Dimensions rather than one so what I mean by that is what what is the function of the verticality the deep and superficial verticality if ascending and descending product predictions is compensated for or counted for by the horizontal axis in this picture in this diagram yeah one answer is that the the the histological um verticality let's just say the fact that there's multiple layers of tissue gives you a really nice packet or motif that has um both the units that you need the MU and the E so it has the two pieces that you need so you could have your workshop with all the with just a bunch of mu's and E's and then just compose it however you needed it um without caring if you're reusing motifs but if we construct a motif like a cortical column that that's able to um do what it does by itself and also um contribute to more nested models just through a diagram and combining them like this it just makes a scalable reusable motif um that's developmentally plausible and actually has support in neurobiology but then Javier isn't the the horizontality here mustn't think about different areas in the brain that could also be seen as a hierarchy you know vertical hierarchy between you know a lower area like you know V1 V2 and V4 in the visual system so we will have you know both verticality within uh you know each of this area and you know a more General connection between areas in which you you know you have a more and more complex uh environment message yes you could coarse grain and you could um look at other neural systems in terms of message passing and hierarchical models go for it heavier yeah um connecting back with a thousand uh brain's theory of Hawkins wouldn't disconnect as well with the fact that you know Hawkins says that you have you don't have this one model you have all these columns and micro columns like creating lots of different models and then you have this sort of voting system that is connected is related to the horizontal Connections in

which the different columns are influencing each other is that a sort of way of explaining this you know vertical versus horizontal so maybe the horizontal sort of voting or influencing uh the models created by the different columns or micro columns yeah that's a great a great um comment one paper that kind of approaches that is this Fame in the brain Fame in the predictive brain um Dennett had this Fame in the brain meme and and uh this paper argues that basically like the the unit the unity of consciousness is like kind of like underlying it there's this subconscious process of selection amongst those models I mean again this is this this is a is running fast and loose with the map in the territory because we're talking about a system of interest but we're obviously talking about the acting funtology but like if um 10 counterfactuals are imagined each one of them in deliberation might be transiently famous and and explicitly considered like you know I could go to coffee shop A I could go to B I could go to Z but then for things that are experienced that can be understood as like the outcome of this election or selection process hmm so it's like the winning the winning hypothesis is what is you know in the spotlight of Consciousness and or awareness at a given time Ali yeah and also aside from that saliency map uh that uh I mentioned before I don't believe uh the representation of the nested hierarchy uh is uh is about I mean a necessarily um uh it's not necessarily spatially distributed or I mean horizontally distributed uh representation it's more like uh the unpacking of a single uh let's say Tangled uh neurons that's unpacking that figure so the real physical or um spatially distributed hierarchy is the one we we have uh uh the the vertical one in which the clusters of different types of neurons are packaged into uh I mean vertical columns so and and that's also I believe uh so Darius asked what's the uh computational justification for um such an organization well that kind of packaging can be efficient if we consider uh the different types of neuron with a different length of axons for each neuron so uh so each of those types of neurons can be assigned with specific tasks or can be more efficient in doing some tasks other than the others so uh yeah I believe that's also one of the reasons for the vertical distribution of those types of neurons but again that nested hierarchy is just schematic representation or unpacking of a single Tangled kind of neurons and not necessarily it doesn't refer to any spatially distributed hierarchy yeah also we can think back to the neats and the scruffies I mean the extreme scruffy position is every synapse needs its own framework and University department and software package on the other end you have the extreme neat framework and and so real work exists on that Continuum and that's part of the unfolding discussion and one can only imagine how much neuroimaging and and molecular histology goes into the development of these really elegant computational um

models and then people take them in different ways some people take these in abstract or generalize them even more and aren't as worried about whether it actually adheres to the structure of the brain um like you know what if we had two cortices on top of each other and we had this and we had that um someone else might want to um make it more directly connected to the measurements they were taking in their lab so again chapter five is is a it's a five switch from chapter four because really is getting into the empirical and the specific systems of Interest that's going to come back again in chapter six when we talk about the recipe and um chapter nine when we talk about integration with data but that's kind of the the breadth that that we're in between on one hand the more theoretical work like free energy principle and active inference in principle and on the other hand these are specific models and what do you call somebody who makes an evidence-based critique of this model a computational neuroscience student or researcher I mean that that's all there is to that field looking at how other people have um integrated the information and made models and then offering um positive developments and critiques beyond that so it's kind of interesting after all like the math and and philosophy that it can feel like in the earlier chapters here which is like dumped squarely into empirical neuroscience and the connections um just signaled by a new chapter just so that we see it before we like write down our questions and and hopefully um you know people can add as much discourse it's super super helpful to vote or add um your own thoughts and curate the answers of others um in the coming week but basically here's what's going to be upcoming we're going to go from looking at the to descending predictions playing a role in motor control proprioception proprioceptive information is going to come in and then in the ventral and dorsal horns but basically in this cross section of the mammalian spinal cord the descending predictions are going to meet with the proprioceptive incoming information and that's going to result in kind of this differential guided motor activity selection like I want my arm to be here it is where I want to be it's not moving move the set point now the r moves then going back up to the cortical column we're going to now Trace another projection so not just a projection to the spinal cord but also cortical layer 5 targeting some deep brain structures like the stratum and other aspects basal ganglia very interesting very cool dopaminergic and and brain regions several topics and cognitive phenomena are discussed that the midbrain regions are important for here in figure 5.4 we also talk about policy selection and the direct and the indirect Pathways of policy selection one of which the indirect is basically a pass-through on habit and the other the direct pathway involving an expected free energy calculation so thus updating prior probabilities on policies habit updating that

according to the expected free energy of policies which is to say their epistemic and pragmatic value so this is kind of a type 1 type 2 or like a habit pass-through versus deliberative decision making distinction table 5.1 provides some empirical work connecting different computational variables to neurotransmitters of course neurotransmitters do different things in different brain regions different synapses different contexts this is the neat and Scruffy again we could have a framework for every single specific neurotransmitter or we could just say all neurotransmitters are the same and this is somewhere in the middle this is saying we can actually understand some some uh uniquely valuable insights and unique predictions from associating different neurotransmitters in certain contexts with different computational roles 5.6 comes to this hybrid theme which is about connecting the continuous and the discrete hierarchies it's a total major theme of the textbook though we are often distinguishing between discrete and continuous State spaces and also a special focus on the continuous and discrete treatment of as with another modeling it's it's Revisited multiple times that you can make Hybrid models for example with continuous modeling at a lower level and category Oracle modeling at a higher level um finally in the closing pages they stitch together all three of the systems that were introduced so the descending signaling coming from uh the cortical column on one hand going down to the motor selection unit however also going down to the behavioral planning and selection components and in fact those behavioral selection and planning components then re-entering into the cortical so that the action can be decided so just one example of uh compositional cognitive cartography and bringing together different modeled subsystems of Interest and using active inference to integrate diverse cognitive phenomena and that's chapter five any last comments well next week will be our uh it's the Symposium so hopefully people can view it or or re-watch it then uh it'll be our final discussion on and the following week will be an unrecorded session for for like feedback um we really need everyone truly so we will be especially interested to hear everyone's feedback with this Anonymous feedback form for people's um suggestions on how to develop and improve there's many many notes so it it for those who want to stay active and and leverage their contribution reading what people have already suggested which is quite copious and then curating or adding it and then maybe even finding a place where like you want to continue to get more involved in the coming months and years then we'll uh take some weeks off and then restart the second half of cohort four with chapter six meanwhile start cohort five from the beginning of the book and uh the work goes on thank you everyone see you next time thank you bye thank you